

Crowd
state

Robot
observations

Reconstruction
methods

Output

Evaluation



x_t



y_t, Ω_t

■ Senseiver-A	y_t
■ Senseiver-G (ours)	$y_{t-15:t}$
■ DINCAE*	$y_{t-1:t+1}$
■ 4DVarNet	200 s
■ 4DVarNet (aug. head)*	200 s
■ EnKF*	$y_{1:t}$

* also predicts its uncertainty $\hat{\sigma}_t$



$\hat{x}_t$



$\hat{\sigma}_t$

accuracy	RMSE
uncertainty	CRPS
	spread/RMSE
speed	ms per frame
on unobserved walkable cells	

ground truth (test days)